

Household Energy Inequality

What this work contributes

The Journal of Cleaner Production article studies estimated direct energy use across 3,637 Chinese households in 85 cities, using a 2014 survey. Chenrui Xu is a coauthor credited with data curation, resources and validation. The publisher PDF remains unchanged.

How to read the findings

Energy quantities are inferred from activities and equipment assumptions, rather than measured household electricity or a complete life-cycle footprint. The typologies and associations do not estimate causal savings from an intervention. Historical data and assumptions limit generalization.

Numerical and methodological cautions

On original PDF page 11, the stated 175k versus 99k and 78k versus 48k comparisons correspond to approximately 76.8% and 62.5% higher values. They are approximately 177% and 163% of the comparison values, not that much higher. These percentages are not repeated in the website description. Standardization alone does not establish homoscedasticity; this review did not reproduce the clustering implementation.

Source and status

Original article, DOI 10.1016/j.jclepro.2024.142870, especially PDF pages 3-4 and 11-12. This is a retrospective portfolio reading note, not a journal-issued correction or a new analysis.

How Universities Introduce Proof

What this work contributes

Alexander Bell, Pablo Ortuno Floria, Andrew Shaw and Chenrui Xu jointly examined introductory proof curricula and student perspectives. The 86-page report combines curriculum coding, questionnaires and descriptive analysis. Individual task allocation has not been established beyond joint authorship.

Corrections to the historical report

Original PDF page 44 states 79.8% for a category whose preceding table records 53 of 76 responses; that fraction is 69.7%. Course counts also disagree: pages 25-26 use 21 and 20 totals, while the categories on page 68 sum to 20. A revised total requires reconciliation with the coding records; no replacement total is asserted here. A category containing 9 of 21 is the largest category, not a majority.

Interpretation

The student groups are unequal and the design is descriptive and cross-sectional. Group differences cannot establish that a course caused a change in attitudes, nor do they constitute a before-and-after study of the same students. Read causal wording in the conclusion as an exploratory hypothesis.

Source and status

Original 2024 team report, PDF pages 3, 12-24, 25-26, 43-44 and 68-71. The archived PDF retains its historical wording. This 2026 note supplies corrections and limits; it does not certify a reconciled full revision.

Seasonal and Spatial Weather Models

What this work contributes

An individual R analysis combining seasonal harmonic terms, geographical covariates, permutation experiments and leave-one-station-out prediction. Holding out whole stations is a useful test of spatial transfer within the course dataset.

Terminology and scale

The DS score is the **Dawid-Sebastiani score**, not the “Decomposition Score” named on original PDF page 16. The response is the square root of monthly mean precipitation. Reported errors therefore require interpretation on that transformed scale; they are not precipitation errors in millimetres.

Validation and inference

Leave-one-station-out prediction concerns unseen stations, not forecasting future years. Permutation inference also relies on exchangeability; arbitrary label permutation does not by itself account for temporal dependence in weather. A zero exceedance count among 1,000 permutations is not proof of a true p-value of zero. The historical report already includes Monte Carlo uncertainty.

Source and status

Original report dated 21 April 2023, PDF pages 4-7, 11, 16 and 19, with archived R Markdown and helper functions. The work uses course-provided StatCompLab material. Historical outputs have been read and checked for scope, not rerun in a frozen software environment.

Hotel Booking Cancellations

What this work contributes

Yu Zhang and Chenrui Xu compared statistical-learning models using 119,210 original reservation records. The work connects cleaning and exploration with logistic regression, regularization and random forests. The archived notebook includes a train/test split and training-set grid search.

Corrections to the historical report

Dividing a predictor by 10 (PDF page 11) is linear scaling. Logistic regression does not require normally distributed predictors (page 12). The random-forest parameter description on page 13 incorrectly includes `logisticregression_C`; the available forest search varies the number of trees, depth and minimum split size.

Evaluation limits

Historical scores have not been reproduced in this review. The notebook refers to an unavailable intermediate `hotel_3.csv` and contains unrelated cells, so a complete raw-data-to-report execution chain is not yet established. A deployment claim also requires defining the prediction time and checking that every feature is available then. A random split does not establish future operating performance.

Source and status

Original team report dated 22 April 2024, especially pages 9-15, and archived `report_4.ipynb`. Associations and proposed business actions are not evidence of reduced cancellations or increased revenue. This note corrects interpretation without replacing the historical results.

Cognitive Scores in Older Adults

What this work contributes

Yujie Chu and Chenrui Xu explored cognitive-score associations using easySHARE, comparing linear and polynomial regression with ridge, lasso and elastic net. The website provides a summary rather than the original report containing participant-level output.

Why no performance headline is given

Target-informed feature screening and transformation fitting occurred before the train/test split. Polynomial degree was then selected by inspecting test errors. The saved test results therefore do not represent an untouched final evaluation.

What a validated revision requires

Hold out the final evaluation data first, place learned transformations and supervised screening inside training folds, and choose model complexity through training-only cross-validation. Check whether repeated participants or survey waves require grouped splitting. Evaluate the chosen procedure once on the held-out data.

Interpretation and source

The target is a recorded cognitive score, not a validated clinical dementia diagnosis. Observational prediction does not establish preventive or causal effects. Source: original team report dated 14 March 2024 and project1 (2).ipynb, especially feature selection, splitting and polynomial-degree cells. This note reports methodological limits; no new clinical or predictive result is claimed.

Dynamics and Epidemic Systems

What this work contributes

The two coursework reports connect Jacobian and eigenvalue analysis, numerical integration, phase portraits and prescribed epidemic scenarios. The public dynamics file is a labelled excerpt; the epidemic file is the later corrected version.

Stability and numerical accuracy

In the dynamics report, $a = +2$ and $a = -2$ give repeated eigenvalues, so the strict inequalities shown at the boundary need separate treatment. At $a = 0$ the displayed oscillator has conserved x^2 plus y^2 , which establishes a centre. Purely imaginary linearized eigenvalues alone are not a general nonlinear stability proof.

What the excerpt does and does not fix

The omitted exercise called a fifth-order Taylor polynomial an exact solution. Removing that exercise improves the selected sample but does not repair the historical numerical-error experiment. In the epidemic report, a sampled maximum from adaptive solver output is not an accurately located peak event. Conservation of total population by Euler updates does not imply nonnegative components for every step size.

Source and status

Dynamics: December 2022 report, original pages 1-3 and 18-22. Epidemic simulation: 31 March 2023 report, pages 5-7 and 14-15, with D. J. Higham and H. Yorston course attribution. Parameter scenarios illustrate specified models; they are not empirical estimates of intervention effects.

PISA: Descriptive Comparisons

What this work contributes

Chenrui Xu contributed descriptive analysis and visualization; Yu Zhang contributed regression modelling. The course project compared samples from PISA 2015 and 2018 across the UK nations and selected European countries. The public attachment is now a concise project brief.

Why the full report is not the main attachment

The original conclusion contradicts some stored regression outputs. Its final region plot contains an NA category; the available R Markdown repeats a factor recoding after labels have already changed. The source version and rendering chain need to be reconciled before issuing a revised full report.

Scope of inference

The main analysis uses one plausible value and does not implement the full survey-weight, replicate-weight and multiple-plausible-value procedure. HC3 robust standard errors appear in the code, but they do not replace those survey-design components. Treat the work as sample exploration, not definitive national ranking, population trend testing or policy-effect estimation.

Source and status

Original team report dated 30 November 2023, pages 1, 15-17 and 22-23, with report.Rmd. The archived source is retained. Neither the coding problem nor the survey-inference limitations have been resolved by substituting a brief; the brief accurately summarizes the supported contribution.

Investment Style During COVID-19

What this work contributes

Xinyu Jiang, Zhuoer Li and Chenrui Xu are equal-contributing authors of a 2021 proceedings paper applying the Fama-French five-factor model to an apparel-sector portfolio before and during COVID-19. The website records this early collaborative work and links the unchanged publisher version.

Retrospective statistical reading

The tables on original PDF page 4 print market-factor p-values as 1.200 and 3.805, outside the valid probability range. The intended values cannot be recovered reliably from the reviewed material. The discussion also describes 18.23 to 25.20 as an increase in the factor, although these entries are t-statistics; the corresponding coefficients are 1.155 and 1.030.

What follows from a two-period comparison

A coefficient being significant in one period and not in another does not itself show a statistically significant change. That requires a direct difference, interaction or structural-change test. The archived regressions do not establish investor preferences, a causal pandemic effect or a profitable investment strategy.

Source and status

ICPRSS 2021 proceedings, DOI 10.2991/assehr.k.211020.338, especially PDF pages 3-6. This retrospective reading note is not a publisher correction. No guessed replacement p-values or newly estimated results are supplied.

Financial Forecasting Implementations

What this work contributes

The Berkeley capstone is team work. The later public-data extension provides compact PyTorch model implementations, chronological splitting, configuration-based training and saved experiment outputs across six assets. These implementations are adaptations, not certified reproductions of the original architecture papers.

Metric correction required

The code standardizes Close targets and evaluates predictions without inverse transformation. Consequently RMSE is in standardized units and MAPE is not a price-percentage error. The directional score measures agreement between successive steps within the future path, not a trading return or the direction from today to the final horizon.

Requirements for a stronger benchmark

Compute price-based metrics after inverse transformation, define each horizon and directional target, include a last-price baseline, fix the download interval, save predictions and targets, and repeat stochastic training where a ranking is claimed. The existing end:null configuration changes the data range over time. Training-set scaler fitting is present; this review did not find a basis for a blanket claim of scaling leakage.

Source and status

Reviewed repository commit `b7827f99b098f581dcfa06e3939c5b39286aaa34`, especially `scripts/dataset.py` and `scripts/train.py`. No original final capstone PDF was established. Existing outputs were inspected, not rerun or corrected; the current card describes implemented methods rather than validated model superiority.

Shared-Mobility Pricing and Matching

What this work contributes

The original Waymodidi policy is team coursework. The later extension adds an event-driven simulator, a conversion-response model and approximate dynamic programming for pricing, while retaining team matching and inherited area prices. The technical writeup is six pages, not the five stated in the README.

Corrections and evaluation limits

The saved course validation and later replay do not establish performance for the same frozen policy version; their difference cannot be assigned solely to simulator bias. Conversion uses a fixed seed, but baseline pricing uses a separate unseeded NumPy random source. Reproducibility of the entire comparison is therefore incomplete.

Demand and model scope

The 11,788 arrivals occupy six one-hour windows: about 32.7 arrivals per minute. This contradicts the README claim that training intensity is an order of magnitude below roughly 28 arrivals per minute. Response fitting and replay also reuse training data. Approximate queue dynamics, online price blending and extrapolation limit claims of optimality or external policy value.

Source and status

Reviewed commit 653160fc17db07d35fb249f4beb29c2229f1f27e. This note corrects how the unchanged archived writeup should be read. Fixing all random sources, binding results to commits and using independent scenarios with repeated runs remain prerequisites for strong performance claims.